ROJECT: LearnHUB — Complete Online Learning and Education Management Platform

1. YOUR ROLE AND MAIN OBJECTIVE

Act as a senior full-stack software engineer, software architect, UI/UX designer, database architect, cybersecurity engineer, and QA engineer.

Your task is to design, develop, test, and prepare for deployment a complete, modern, professional, responsive, secure, and scalable online learning platform named LearnHUB.

Build the entire application from scratch using the MERN stack:

MongoDB — database

Express.js — backend API framework

React.js — frontend

Node.js — backend runtime

Use JavaScript or TypeScript consistently throughout the application. Prefer TypeScript if the existing environment supports it without unnecessary complexity.

Do not stop after creating a homepage, static UI, sample dashboard, or basic prototype. Build a functional full-stack application with connected frontend pages, backend APIs, database models, authentication, role-based authorization, file management, notifications, scheduling, testing, and deployment documentation.

The application must be suitable for real users after the required environment variables, storage services, email services, and deployment settings have been configured.

Important execution rule: Do not merely describe what should be built. Inspect the workspace, create the actual project files, implement the features, run the available checks, and fix errors. Continue working until the complete achievable MVP is implemented.

If a feature requires external credentials or paid services, implement its integration interface, secure configuration, and graceful fallback. Never pretend that an external service is connected when it is not.

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2. PRODUCT VISION

LearnHUB is a comprehensive online learning platform that helps students learn languages, technology, programming, digital marketing, academic subjects, professional skills, and other educational topics.

The platform should support different learning formats, including:

Live online classes

Teacher-led courses

Scheduled lessons

Recorded educational videos

PDF and document-based learning

Learning resources and downloadable materials

Assignments and assessments

Student progress tracking

Future certificates and subscriptions

LearnHUB must support different categories of learning rather than focusing on language learning alone.

Use a modern, scalable architecture that allows additional learning categories and features to be introduced later.

The product should be inspired by the general usability and learning-management concepts of modern education platforms, but it must have its own identity, design, source code, content, database, and implementation.

Do not copy proprietary source code, branding, text, images, or protected assets from another website.

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3. NON-NEGOTIABLE BUSINESS RULES

These rules are mandatory and must be enforced by the backend, not just by hiding frontend buttons.

Rule 1: Hub Admin Controls Teacher Hiring

Only an authorized Hub Admin can:

Review teacher applications

Approve or reject teacher applications

Create or activate teacher accounts

Suspend or reactivate teachers

Assign teachers to classes

Change teacher assignments

Teachers must never have permission to hire, approve, create, or manage other teachers.

Rule 2: Hub Admin Controls Student Placement

Students can register, submit learning goals, select interests, indicate their current skill levels, and request learning opportunities.

However, students must not independently assign themselves to a class or teacher.

The Hub Admin reviews student information and assigns each student to the appropriate course, class, and teacher.

Only authorized Hub Admin actions can create or change these assignments.

Rule 3: Teachers Teach Their Assigned Classes

Teachers can access only the classes and students assigned to them, subject to the permissions explicitly granted by the platform.

Teachers can:

View assigned classes

View their class schedules

Access assigned students' relevant learning information

Upload and manage permitted learning resources

Conduct scheduled lessons

Add lesson descriptions and learning materials

Create or grade permitted assignments

Record attendance and lesson completion

Provide feedback

Track progress for assigned students

Receive notifications about upcoming classes

Teachers cannot:

Hire teachers

Approve teacher applications

Assign students to classes

Change their own class assignments

Access unrelated classes

Access private administrative records

Elevate their own permissions

Rule 4: Students Access Assigned Learning Only

Students can:

Manage their permitted profile information

Submit learning preferences

View their assigned courses and classes

View their assigned teachers

View class schedules

Access authorized Google Meet links

Watch authorized videos

View or download permitted learning files

Submit assignments

Take available assessments

View their learning progress

Receive notifications

Contact support

Students cannot independently place themselves in a class, change their assigned teacher, or access another student's private information.

Rule 5: Enforce Permissions on Every Protected Backend Endpoint

Implement server-side role-based access control and resource ownership checks.

Never rely only on frontend route guards, hidden buttons, or client-side role values.

Validate the user's identity, role, and permission to access the requested resource for every protected operation.

Rule 6: Real Data and Real Functionality

Do not use static arrays as the permanent data source for application features.

Use MongoDB-backed data for users, courses, classes, assignments, schedules, notifications, learning resources, and other persistent information.

Seed data may be used for development, but the production application must work with actual database records.

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4. BRAND IDENTITY AND UI/UX DESIGN

Create a distinctive and professional brand for LearnHUB.

Brand personality

Modern

Educational

Trustworthy

Innovative

Accessible

Student-centered

Professional

Technology-oriented

Visual direction

Use a clean, modern educational SaaS design.

Suggested visual palette:

Primary: Deep navy blue

Secondary: Royal blue

Accent: Teal or cyan

Background: White and very light gray

Text: Dark slate

Success: Accessible green

Warning: Amber

Error: Accessible red

Use the colors consistently through design tokens or CSS variables.

Avoid excessive gradients, unnecessary animations, oversized cards, cluttered dashboards, and generic AI-generated landing page layouts.

Create a coherent design system including:

Typography

Color tokens

Spacing scale

Buttons

Inputs

Cards

Dialogs

Dropdowns

Navigation

Tables

Badges

Alerts

Toast notifications

Loading states

Empty states

Error states

Use appropriate icons from a maintained icon library.

Use subtle transitions, clear hierarchy, readable typography, and consistent spacing.

Create an original LearnHUB logo using text and a simple educational or knowledge-related symbol. If no logo asset exists, build a clean SVG or CSS-based logo.

Do not depend on inaccessible remote images for essential UI elements.

Responsive design

The entire application must work on:

Mobile phones

Tablets

Laptops

Desktop screens

Create responsive navigation, mobile-friendly forms, accessible tables, adaptable dashboard cards, and usable touch targets.

Accessibility

Follow practical WCAG accessibility principles:

Semantic HTML

Keyboard navigation

Visible focus states

Proper form labels

Sufficient color contrast

Accessible modal dialogs

Helpful validation messages

Appropriate alternative text

Screen-reader-friendly controls

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5. LANGUAGE SUPPORT

The primary interface language should be Amharic.

Use natural, clear, professional Amharic for user-facing labels, instructions, dashboard navigation, forms, notifications, and empty states.

Also support English as a secondary interface language.

Implement a proper internationalization system rather than scattering translated strings throughout the code.

Requirements:

Language selector

Amharic translations

English translations

Persistent language preference

Consistent translated navigation

Translated form validation messages where practical

Correct font rendering

Support for Amharic Unicode characters

Use an appropriate Amharic-capable font with a reliable fallback.

For Amharic layouts, verify that text alignment and layout remain readable. Support RTL layouts if additional RTL languages are introduced later, but do not force Amharic into RTL because Amharic is written left to right.

Use consistent terminology throughout the application.

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6. PUBLIC WEBSITE PAGES

Build a complete public-facing website.

6.1 Homepage

Create a polished homepage with:

Navigation bar

LearnHUB logo

Home

Explore Learning

Learning Categories

How It Works

About

Contact

Sign In

Get Started

Hero section

Strong educational headline

Supporting description

Primary call-to-action

Secondary call-to-action

Learning category cards

How LearnHUB works

Featured learning opportunities

Benefits for students

Benefits for teachers

Platform statistics based on real data when available

Testimonials only when genuine data exists

Frequently asked questions

Final call-to-action

Footer with policies and contact information

Do not invent real user counts, ratings, testimonials, partnerships, or institutional affiliations.

If real data is unavailable, use neutral promotional content rather than fabricated statistics.

6.2 Learning Categories

Create categories such as:

Languages

Web Development

Programming

Computer Science

Digital Marketing

Academic Subjects

Business Skills

Professional Development

Creative Skills

Other Learning

Allow administrators to add, edit, reorder, activate, and deactivate categories.

6.3 About Page

Explain the mission, vision, educational purpose, and values of LearnHUB.

6.4 Contact Page

Include a contact form with backend validation and secure submission.

Store inquiries in MongoDB and provide an administrative interface for managing them.

Protect against spam and abuse using rate limits and input validation.

6.5 FAQ Page

Provide searchable frequently asked questions covering registration, teacher approval, student placement, classes, learning resources, scheduling, and support.

Allow administrators to manage FAQ entries.

6.6 Legal Pages

Create editable or maintainable pages for:

Privacy Policy

Terms of Service

Cookie Policy, if applicable

Acceptable Use Policy

Make clear that the initial legal text is a template requiring appropriate review before production use.

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7. AUTHENTICATION AND ACCOUNT MANAGEMENT

Implement secure authentication.

Required features

Student registration

Teacher application and registration workflow

Login

Logout

Email verification

Forgot password

Secure password reset

Session management

Profile management

Account status management

Password change

Authentication error handling

Protected routes

Role-based dashboards

Teacher registration workflow

Teacher registration must create a teacher application or pending teacher account.

The teacher must not receive active teaching permissions until an authorized Hub Admin approves the application.

After approval, the teacher can access the teacher dashboard and perform permitted teaching tasks.

The system must never let an applicant grant themselves an approved role by changing a request payload.

Student registration workflow

Students can create accounts and submit:

Full name

Email

Password

Preferred language

Learning interests

Learning goals

Current skill level

Preferred schedule, if applicable

Optional additional information

Students can update permitted profile fields later.

Submitting a learning preference does not create a class assignment.

Authentication security

Hash passwords with Argon2 or bcrypt using secure parameters.

Validate all input on the backend.

Use secure session or token handling.

Prefer HttpOnly, Secure, SameSite cookies where appropriate.

If JWTs are used, implement safe token storage and refresh/revocation strategy.

Never store plaintext passwords.

Never expose secrets in frontend code.

Add rate limiting to login and password reset.

Use generic authentication error messages where appropriate.

Protect against brute-force attacks.

Verify email ownership before enabling sensitive email-dependent actions.

Invalidate or rotate sessions appropriately after password changes and security-sensitive actions.

Create a secure initial admin provisioning mechanism through environment variables or a documented one-time setup command.

Never expose an unrestricted public admin registration endpoint.

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8. USER ROLES AND PERMISSIONS

Implement the following roles:

Hub Admin

Teacher

Student

Support Staff, with limited permissions if enabled

Use a centralized permission system.

Hub Admin permissions

The Hub Admin can:

View administrative dashboard

Review platform statistics

Manage teacher applications

Approve or reject teachers

Activate or suspend teacher accounts

Review student registrations and learning preferences

Assign students to courses and classes

Assign teachers to classes

Create and manage courses

Create and manage classes

Create schedules

Manage Google Meet links

Manage learning categories

Manage learning resources and file access

Manage announcements

View attendance and progress reports

Manage FAQ and contact inquiries

Manage appropriate platform settings

Review audit logs

Manage account status

View operational reports

Administrative permissions must be protected on the backend.

Teacher permissions

Teachers can:

Access their dashboard

View assigned classes

View their timetable

View permitted student information

Manage permitted lesson materials

Upload class resources

View and manage permitted assignments

Record attendance

Record lesson completion

Provide student feedback

View permitted student progress

Receive reminders and notifications

Teachers cannot access admin-only management features.

Student permissions

Students can:

Access their dashboard

View assigned courses and classes

View assigned teacher information

View schedules and authorized meeting links

Access authorized resources

Submit assignments

Complete assessments

View their own progress

Receive announcements and reminders

Update permitted profile information

Support Staff permissions

If implemented, Support Staff can handle assigned support inquiries and view only the information necessary for support.

Support Staff must not have unrestricted administrative, teacher-hiring, student-placement, or permission-management capabilities.

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9. HUB ADMIN DASHBOARD

Build a complete administrative dashboard.

Dashboard overview

Display real database-backed metrics such as:

Total students

Active students

Pending student placement

Total teachers

Pending teacher applications

Active courses

Active classes

Upcoming classes

Completed sessions

Pending support inquiries

Handle empty databases gracefully.

Teacher management

Create a complete teacher management module with:

Teacher application list

Application details

Qualifications and experience fields

Application status

Approve button

Reject button

Rejection reason

Account activation and suspension

Teacher profile

Teacher availability

Assigned classes

Teaching schedule

Relevant history

Record administrative decisions in an audit log.

Student management

Create:

Student list

Search and filtering

Student profile

Learning goals

Learning interests

Skill level

Placement status

Current assignments

Learning history

Account status

Student assignment workflow

The Hub Admin must be able to select a student, choose a course or class, select an available teacher, confirm the assignment, and save the assignment.

Validate teacher availability and class capacity when applicable.

Prevent accidental duplicate assignments.

Provide confirmation dialogs for sensitive actions.

Course management

The Hub Admin can:

Create courses

Edit courses

Archive or deactivate courses

Assign categories

Set course descriptions

Set learning objectives

Set skill level

Configure publication status

Organize course content

Link resources

Define class requirements

Support draft and published states.

Class management

The Hub Admin can:

Create classes

Edit classes

Assign courses

Assign teachers

Add or remove students

Set capacity

Configure recurring or one-time schedules

Set start and end dates

Add meeting links

Manage class status

View attendance

View class resources

Only authorized administrative actions may change student or teacher assignments.

Scheduling management

Create an administrative calendar and schedule editor.

Support:

Date

Start time

End time

Time zone

Teacher

Class

Student group

Session description

Session status

Meeting link

Reminder configuration

Prevent overlapping schedules for the same teacher where appropriate.

Use a consistent time zone strategy. Store timestamps in UTC and display them in the configured local time zone.

Allow an appropriate platform time zone to be configured through environment settings.

Resource management

Administrators can upload, inspect, categorize, and manage learning resources according to permissions.

Announcements

Allow administrators to publish announcements for:

All students

All teachers

Specific classes

Selected authorized groups

Prevent recipients from seeing private information about unrelated users.

Reports

Provide practical reports for:

Student placement

Teacher assignments

Class attendance

Lesson completion

Student progress

Resource usage

Upcoming classes

Teacher workload

Use real database information.

Include pagination, filters, and export options where appropriate.

Audit logs

Record important administrative operations, including:

Teacher approval and rejection

Student placement and reassignment

Teacher assignment changes

Class schedule changes

Account suspension

Permission-sensitive changes

Resource access changes

Audit records must not be editable by ordinary users.

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10. TEACHER DASHBOARD

Every approved teacher must have a dedicated dashboard.

Overview

Show:

Assigned classes

Upcoming lessons

Today's teaching schedule

Total assigned students

Pending assignments to grade, if applicable

Recent notifications

Recent teaching activity

My Classes

Teachers can view their assigned classes and permitted class details.

For each class, show:

Class name

Course

Students

Schedule

Meeting link

Learning materials

Lesson history

Assignments

Attendance

My Schedule

Build a calendar and list view.

Allow teachers to view upcoming and completed sessions.

Teachers must not be able to change their assigned class, change the official teacher assignment, or edit protected administrative schedules.

If schedule-change requests are supported, teachers may submit requests for administrative review.

Lesson management

Allow teachers to:

Add lesson titles and descriptions

Add learning objectives

Upload permitted lesson materials

Mark lessons as completed

Record attendance

Add teaching notes visible only to authorized roles

Create assignments when permitted

Review student submissions

Provide feedback

Record relevant changes and timestamps.

Student progress

Allow teachers to view progress only for their assigned students.

Display appropriate information such as:

Attendance

Assignment completion

Assessment results

Lesson completion

Teacher feedback

Learning progress

Protect sensitive information and avoid exposing one student's private records to another.

Notifications

Notify teachers about:

New class assignments

Upcoming lessons

Schedule changes

Student submissions

Administrative announcements

Relevant class updates

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11. STUDENT DASHBOARD

Create a welcoming and easy-to-use student dashboard.

Overview

Display:

Welcome message

Assigned courses

Current classes

Next lesson

Assigned teacher

Upcoming schedule

Recent learning resources

Pending assignments

Recent notifications

Learning progress

My Learning

Students can see their officially assigned learning opportunities.

Each course should display:

Course name

Course description

Teacher

Learning objectives

Class schedule

Course status

Authorized lessons

Authorized resources

Progress information

My Classes

Display:

Class name

Teacher

Schedule

Meeting link

Class status

Relevant announcements

Available learning materials

Students must only see classes they are authorized to access.

My Resources

Allow students to view or download files associated with their assigned classes or courses.

Check resource authorization on every access.

My Assignments

Students can:

View assignment instructions

View due dates

Submit supported file types or text answers

Review submission status

Read teacher feedback

View grades when released

Prevent unauthorized modification of other students' submissions.

My Progress

Display progress based on real data, including:

Completed lessons

Attendance

Assignment completion

Assessment results

Course progress

Do not generate fake progress percentages.

Explain the progress calculation and handle courses without completed activities.

Learning preferences

Students may update permitted preferences and goals.

Changing preferences must not automatically change their class or teacher assignment.

The Hub Admin reviews and manages placement.

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12. COURSES, LESSONS, AND LEARNING CONTENT

Create a flexible learning content system.

Course structure

A course may contain:

Course information

Learning objectives

Sections or modules

Lessons

Learning resources

Assignments

Assessments

Scheduled sessions

Progress records

Lesson types

Support:

Text lesson

Video lesson

PDF lesson

Downloadable resource

Live class

External learning resource

Mixed-media lesson

Video learning

Support authorized video playback.

Use an appropriate storage or video-hosting integration.

Do not load large video files directly into MongoDB.

For production, prefer private object storage or a dedicated video service with authorization and appropriate streaming support.

If signed URLs are used, make them short-lived and issue them only after authorization.

Course visibility

Support draft, published, archived, and inactive states where appropriate.

Students must not access unpublished content unless explicitly authorized.

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13. FILE STORAGE AND RESOURCE MANAGEMENT

This is a core requirement.

Implement a secure file storage system for learning materials.

Supported file categories

Support appropriate combinations of:

MP4 and other approved video formats

PDF

DOC and DOCX

PPT and PPTX

XLS and XLSX

JPEG and PNG

WebP

MP3 and approved audio formats

TXT

ZIP, only if explicitly allowed and safely handled

Other approved educational file types

Storage architecture

Use object storage for actual file contents.

Possible providers include Amazon S3-compatible storage, Cloudflare R2, or another compatible object storage service.

Use MongoDB to store resource metadata, including:

Original filename

Stored object key

MIME type

File size

Uploader

Upload timestamp

Associated course

Associated class

Visibility

Resource category

Processing status, if applicable

Do not store large file contents as MongoDB documents.

Upload workflow

Implement:

User authentication.

Permission validation.

File size validation.

MIME type and extension validation.

Upload authorization.

Safe storage key generation.

Upload to configured storage.

Metadata creation.

Authorized resource listing.

Secure download or viewing.

Enforce configurable upload limits.

Do not trust the client-provided MIME type alone.

Prevent executable or dangerous files from being uploaded as educational documents.

Use a malware-scanning integration if available and document its production configuration.

Access control

Teachers may upload resources only to classes or courses they are authorized to manage.

Students may upload submissions only where permitted.

Students must not gain access to files belonging to unrelated classes.

Administrators can manage resources according to their permissions.

Use private storage by default.

Do not expose private storage credentials or permanent public links.

If external storage credentials are missing, implement a clear configuration error and development-only fallback that does not pretend to be production-ready.

Document the setup in the README.

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14. GOOGLE MEET INTEGRATION

Every scheduled live teaching session should support a Google Meet link.

MVP implementation

Allow the Hub Admin to:

Create a session

Add a valid Google Meet URL

Edit the meeting URL

Assign the session to a class

Set the session date and time

Configure reminders

Validate the URL format and permitted domain.

Display meeting links only to authorized participants.

Students must not see meeting links for unrelated classes.

Teachers must see links for their assigned sessions.

Future automated integration

Design the system so automatic Google Meet creation can be added through the appropriate Google Calendar API and Google authorization flow.

Do not claim automatic meeting creation works unless the required API, credentials, scopes, and authorization have been configured and tested.

If credentials are unavailable, manual Meet links must still work.

Meeting experience

For upcoming sessions, display:

Class title

Teacher

Date

Start time

End time

Meeting link

Countdown or time-until-session indicator, if practical

Show a clear Join Meeting button when the session is available.

Do not claim that users are inside a live meeting simply because they clicked the link.

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15. NOTIFICATION AND REMINDER SYSTEM

Build a persistent notification system.

In-app notifications

Create notifications for:

Teacher approval or rejection

New teacher assignment

New student placement

Class schedule creation

Schedule changes

Upcoming class reminders

New learning resources

Assignment creation

Assignment submission

Teacher feedback

Announcements

Relevant administrative messages

Notification features

Notification list

Unread count

Mark as read

Mark all as read

Notification timestamp

Link to relevant authorized resource

Pagination or lazy loading

Scheduled reminders

Implement a reliable server-side reminder mechanism for upcoming lessons.

The system should:

Identify sessions approaching their start time

Determine eligible recipients

Avoid sending duplicate reminders

Record delivery status

Respect user preferences where applicable

Handle retries and failures safely

Use a configurable reminder interval.

Do not rely on a browser tab remaining open for scheduled notifications.

Email notifications

Create an optional email integration for:

Email verification

Password reset

Teacher application status

Class assignments

Schedule reminders

Important announcements

Use a configurable SMTP provider or another suitable email service.

If credentials are missing, keep the application functional and provide a clear development-mode message.

Never expose email service credentials to the browser.

Push notifications can be added later unless they are straightforward to implement without compromising the core MVP.

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16. ASSIGNMENTS, ASSESSMENTS, AND ATTENDANCE

Implement a practical first version of learning evaluation.

Assignments

Teachers with the appropriate class permissions can create assignments containing:

Title

Description

Instructions

Class

Due date

Allowed submission types

Optional attached resources

Maximum score, if graded

Students can submit text responses or approved files.

Store submission metadata in MongoDB and actual files in configured storage.

Submission management

Teachers can:

View submissions for assigned classes

Mark submissions as reviewed

Enter grades where applicable

Provide feedback

Track missing submissions

Students can view their own submission status and released feedback.

Prevent unauthorized grade changes.

Record relevant changes.

Assessments

Implement a basic quiz or assessment module where practical.

Support objective question types such as multiple-choice questions and appropriate server-side scoring.

Do not expose correct answers to students before an assessment is submitted or released.

Attendance

Teachers can record attendance for their assigned classes.

Possible statuses:

Present

Absent

Late

Excused

Students can view their own attendance history.

Administrators can access authorized attendance reports.

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17. STUDENT PROGRESS AND REPORTING

Create a reliable progress model based on actual learning activity.

Track appropriate events such as:

Lesson completion

Assignment submission

Assessment completion

Attendance

Course milestones

Use server-side validation to prevent students from directly assigning themselves arbitrary progress values.

Where progress is calculated as a percentage, document the formula and handle missing activities safely.

For example, a course progress calculation may consider completed lessons and completed required activities, provided that the rules are consistent and clearly defined.

Display progress visually through accessible progress bars and summary cards.

Create reports that allow authorized administrators to identify:

Students awaiting placement

Students with missed sessions

Upcoming class workloads

Assignment completion trends

Teacher workloads

Course participation

Avoid unnecessary surveillance or collecting information that is not needed for education.

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18. DATABASE DESIGN

Create clear MongoDB/Mongoose models with appropriate indexes, validation, timestamps, and relationships.

Use a consistent naming convention.

At minimum, consider these models:

User

StudentProfile

TeacherProfile

TeacherApplication

LearningCategory

Course

CourseModule

Lesson

Class

ClassEnrollment

TeacherAssignment

ClassSession

LearningResource

Assignment

AssignmentSubmission

Assessment

Question

AssessmentAttempt

AttendanceRecord

ProgressRecord

Notification

Announcement

ContactInquiry

FAQ

AuditLog

PlatformSetting

You may adjust the model structure when a more reliable normalized design is appropriate.

Do not duplicate authoritative assignment information unnecessarily.

Define clear relationships and enforce consistency when records are created, updated, or deleted.

Use pagination for large datasets.

Add indexes for common lookups, including user email, roles, application status, class assignments, session dates, notification ownership, and resource relationships.

Use transactions where necessary and supported, especially for multi-record operations that must remain consistent.

Handle deletion carefully. Prefer archival or soft deletion for records that must remain available for auditing or educational history.

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19. BACKEND API ARCHITECTURE

Create a well-organized Express backend.

Separate concerns into:

Routes

Controllers

Services

Models

Middleware

Validation schemas

Authorization utilities

Error handlers

Configuration

Background jobs

Integration adapters

Tests

Use consistent REST API conventions.

Possible API areas include:

/api/auth

/api/users

/api/students

/api/teachers

/api/teacher-applications

/api/categories

/api/courses

/api/classes

/api/enrollments

/api/teacher-assignments

/api/sessions

/api/resources

/api/assignments

/api/submissions

/api/assessments

/api/attendance

/api/progress

/api/notifications

/api/announcements

/api/reports

/api/contact

/api/faqs

/api/admin

/api/settings

Use clear status codes and structured JSON responses.

Implement:

Central error handling

Request validation

Authentication middleware

Role authorization middleware

Resource ownership checks

Request logging without sensitive data

Rate limiting

Security headers

CORS configuration

Safe pagination

Filtering and sorting validation

Consistent API error responses

Do not return internal stack traces or secret configuration in production responses.

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20. FRONTEND ARCHITECTURE

Build a maintainable React application.

Organize the frontend into reusable components, layouts, pages, hooks, services, types, localization files, and utilities.

Use a consistent routing strategy.

Create separate layouts for:

Public website

Authentication pages

Hub Admin dashboard

Teacher dashboard

Student dashboard

Support dashboard, if enabled

Create reusable components for:

Buttons

Inputs

Selects

Form validation

Data tables

Pagination

Cards

Dialogs

Confirmation modals

File uploads

Resource previews

Notifications

Calendars

Empty states

Loading skeletons

Error states

Toast messages

Connect each page to the backend.

Use a suitable data-fetching and caching approach.

Handle loading, empty, success, and error states.

Protect routes on the frontend for a good user experience, while keeping backend authorization authoritative.

Prevent repeated network requests and handle expired sessions gracefully.

Do not use placeholder buttons that appear functional but do nothing.

Every visible action must either perform its intended function or be clearly marked as unavailable or planned.

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21. SECURITY AND PRIVACY

Treat security as a core feature.

Implement:

Password hashing

Secure authentication

Backend role-based authorization

Resource ownership validation

Input validation and sanitization

Rate limiting

Secure HTTP headers

Configured CORS

Protection against common injection risks

Safe file uploads

Private resource access

Short-lived signed download URLs where appropriate

Secure password reset tokens

Session expiration and revocation

Audit logging

Environment-based secret management

Production error handling

Secure administrative provisioning

Database access restrictions

Backup and restoration documentation

Do not commit secrets to source control.

Create .env.example with placeholder values only.

Do not put private API keys, database credentials, JWT secrets, or storage credentials in frontend environment variables.

Avoid logging passwords, reset tokens, authorization headers, or private file URLs.

Make sure students cannot change their own role by editing requests.

Make sure teachers cannot approve themselves or assign students by calling backend endpoints directly.

Make sure users cannot access other users' records by guessing IDs.

Validate every file and resource request on the server.

Use appropriate privacy practices for student information.

Document production security settings and operational limitations.

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22. ERROR HANDLING AND USER EXPERIENCE

Provide understandable errors throughout the platform.

Examples:

Invalid login

Email already registered

Email not verified

Teacher application pending

Account suspended

Unauthorized resource access

File exceeds upload limit

Unsupported file type

Class is full

Teacher has a schedule conflict

Session not found

Invalid meeting link

Network failure

Database unavailable

Storage configuration missing

Show clear messages in the selected interface language.

Do not expose sensitive implementation details.

Add useful empty states, retry options, confirmation dialogs, and loading indicators.

Avoid silent failures.